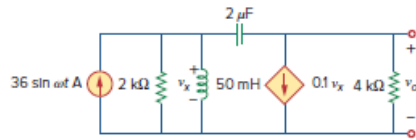


Problem

Use nodal analysis to find v_o in the circuit of Fig. 10.59. Let $\omega = 2 \text{ krad/s}$.

Figure 10.59:



Step-by-step solution

Step 1 of 6

Refer to the circuit diagram in Figure 10.59 in the textbook.

Angular frequency is,

$$\begin{aligned}\omega &= 2 \text{ k rad / sec} \\ &= 2000 \text{ rad/sec}\end{aligned}$$

Convert the circuit into the frequency domain.

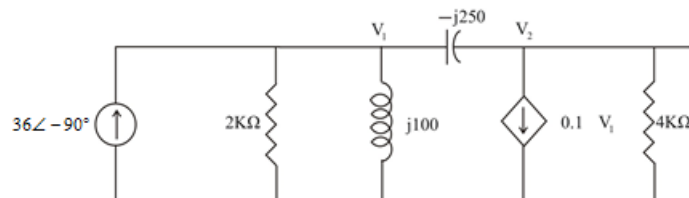
$$\begin{aligned}50 \text{ mH} &\Rightarrow j\omega L \\ &\Rightarrow j2000 \times 50 \times 10^{-3} \\ &\Rightarrow j100\end{aligned}$$

$$\begin{aligned}2 \mu\text{F} &\Rightarrow \frac{1}{j\omega C} \\ &\Rightarrow \frac{1}{j2000 \times 2 \times 10^{-6}} \\ &\Rightarrow -j250\end{aligned}$$

$$36 \sin \omega t \Rightarrow 36 \angle -90^\circ$$

Step 2 of 6

Consider the frequency - domain equivalent circuit below.



Step 3 of 6

Apply KCL at node 1.

$$36\angle -90^\circ = \frac{V_1}{2000} + \frac{V_1}{j100} + \frac{V_1 - V_2}{-j250}$$

$$36\angle -90^\circ = \frac{V_1}{2000} - \frac{jV_1}{100} + \frac{j(V_1 - V_2)}{250}$$

$$36\angle -90^\circ = 0.0005V_1 - j0.01V_1 + j0.004V_1 - j0.004V_2$$

$$36\angle -90^\circ = (0.0005 - j0.006)V_1 - j0.004V_2 \dots\dots (1)$$

Step 4 of 6

Apply KCL at node 2.

$$\frac{V_1 - V_2}{-j250} = 0.1V_1 + \frac{V_2}{4000}$$

$$0 = (0.1 - j0.004)V_1 + (0.00025 + j0.004)V_2$$

$$-(0.1 - j0.004)V_1 = (0.00025 + j0.004)V_2$$

$$(-0.1 + j0.004)V_1 = (0.00025 + j0.004)V_2$$

$$V_1 = \frac{(0.00025 + j0.004)}{(-0.1 + j0.004)} \times V_2$$

$$V_1 = (-8.986 \times 10^{-4} - j0.04)V_2$$

$$V_1 = (-0.0008986 - j0.04)V_2 \dots\dots (2)$$

Step 5 of 6

Equation-(2) substituted in Equation-(1), we get,

$$36\angle -90^\circ = (0.0005 - j0.006) \times (-0.0008986 - j0.04)V_2 - j0.004V_2$$

$$36\angle -90^\circ = (-2.4045 \times 10^{-4} - j1.4608 \times 10^{-5})V_2 - j0.004V_2$$

$$36\angle -90^\circ = (-2.4045 \times 10^{-4} - j1.4608 \times 10^{-5} - j0.004)V_2$$

$$36 = (-2.4045 \times 10^{-4} - j4.01461 \times 10^{-3})V_2$$

$$V_2 = \frac{36\angle -90^\circ}{(-2.4045 \times 10^{-4} - j4.01461 \times 10^{-3})}$$

$$V_2 = 36\angle -90^\circ \times (-14.8656 + j248.199)$$

$$V_2 = 8935.164 + j535.1616 \text{ V}$$

Step 6 of 6

The voltage across load is,

$$V_0 = V_2$$

$$V_0 = 8935.164 + j535.1616 \text{ V}$$

Convert into polar form.

$$V_0 = 8951.17 \angle 3.427^\circ \text{ V}$$

$$V_0 = 8.951 \angle 3.427^\circ \text{ kV}$$

Therefore,

$$V_0(t) = 8.951 \cos(2000t + 3.427^\circ) \text{ kV}$$

$$V_0(t) = 8.951 \sin(2000t + 93.43^\circ) \text{ kV} \quad \text{since} \quad \sin(x \pm 90^\circ) = \pm \cos x$$